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SUMMARY

I am a Computer Science student with a strong interest in data analysis and machine learning-based solutions. I am committed to solving real-world problems through data-driven decision-making and algorithmic modeling. I have hands-on experience working on several academic and personal projects using tools such as R, SQL, TensorFlow, and Keras. I am eager to work in an environment that offers opportunities for learning, collaborative teamwork, and continuous growth in technical skills.

EDUCATION

American International University-Bangladesh (AIUB), Dhaka	Ongoing
Bachelor of Science in Computer Science And Engineering (CSE)	Dhaka
CGPA: 3.37/4.00	

Uttara Model Collage, Dhaka Education Board	2020
Higher Secondary Certificate (HSC)	Dhaka
GPA: 4.75/5.00	

Rupasdi Brindabon High School, Comilla Education Board	2018
Secondary School Certificate (SSC)	Comilla
GPA: 4.50/5.00	

TECHNICAL SKILLS

Programming Languages

C++, R, Java, C#, HTML, JavaScript, Python

Simulation & Design Tools

MATLAB, Emu8086, AutoCAD

Research Tools

TensorFlow, PyTorch, NumPy, Pandas, Keras, Google Colab

Others

MS Word, PowerPoint, Communication Skills

PROJECTS

Clustering Amazon Product Reviews Using TF-IDF, PCA, and Multi-Algorithm Analysis

Developed an unsupervised NLP pipeline to cluster 200,000 Amazon product reviews using TF-IDF, PCA, and three clustering algorithms (K-Means, Hierarchical, DBSCAN). Implemented fallback logic for robust cluster assignment and visualized results using PCA plots. Extracted top terms per cluster and saved annotated datasets for academic reporting.

Handwritten Digit Recognition & Real-time Detection

Developed a CNN-based system using TensorFlow/Keras to recognize handwritten digits from the MNIST dataset and implemented real-time detection using OpenCV with webcam input. Achieved ~98% accuracy and visualized predictions with confidence levels.

Animal Image Classification using KNN

Developed an image classification system using K-Nearest Neighbors (KNN) on a small animal dataset (cats, dogs, pandas). Implemented image preprocessing with OpenCV and performed 5-fold cross-validation to determine the optimal k. Achieved ~43% average accuracy due to small dataset size.

Multi-Class Classification Using a Custom Neural Network on Tabular Data

Implemented a feedforward neural network with 4 hidden layers and dropout to classify tabular data into 3 classes. Achieved 79% test accuracy and evaluated model using precision, recall, and F1-score.

Registration And Login Page

Developed a registration and login page using HTML, with JavaScript form validation to ensure proper user authentication and accurate data entry.

Coaching Management System

Developed a desktop-based system using C# and PL/SQL to manage student profiles, class schedules, teacher information, fee collection, and automated reports.

Online Tutor Management System

Developed an online tutor management system using C#, implementing basic CRUD operations to manage books and user records efficiently.

Car Racing

Car Racing (C++ GUI): Developed a 2D car racing game using C++ and a GUI framework, featuring keyboard controls, score/lap tracking, and basic collision detection.

RESEARCH EXPERIENCE

A Deep Ensemble Approach to Facial Emotion Recognition

Ongoing

This research focuses on building an ensemble framework that combines multiple deep learning models to enhance the accuracy of facial emotion recognition. By leveraging CNN-based architectures, the system aims to classify emotional states from facial expressions with improved reliability. The project contributes to advancements in human-computer interaction and automated affective analysis.

Leaf Guard: Revolutionizing Plant Health with Intelligent Disease Detection

Ongoing

This research explores the development of a machine learning-based system for detecting plant diseases through leaf image analysis. The study involves training classification models using TensorFlow and Google Colab to identify visual symptoms with high accuracy. The objective is to contribute to sustainable agriculture by enabling early diagnosis and reducing crop loss through intelligent intervention.

CERTIFICATIONS

Research Paper Writing Masterclass

CARSIT (Centre for Applied Research in Software & IT)

References

References available upon request.